Laminar Air Flow Cabinet



Category: **Laboratory Instruments & Glassware** Type: **Full SOP** Priority: **High** 3 labs

Covers these lab items: Laminar Air Flow (LAF) Cabinet; Laminar Flow; Laminar flow transfer hood

What it is used for

A laminar air flow cabinet provides filtered, unidirectional airflow over a work surface to reduce particle contamination during suitable laboratory procedures. It is not a substitute for a biological safety cabinet and does not protect the operator from hazardous biological agents, toxic chemicals, or infectious aerosols.

Main parts

- Cabinet enclosure defines the protected work area.
- HEPA filter removes particles from the supplied airflow.
- Blower moves air through the filtration system.
- Work surface supports materials during procedures.
- Front opening provides access to the work zone.
- Airflow grille or diffuser distributes filtered air.
- Control panel operates the blower and fitted lighting.
- Pressure gauge or filter indicator monitors filter condition where provided.

Before you start

1. Only trained laboratory personnel should operate the cabinet for approved procedures.
2. Confirm that the cabinet type is appropriate for the intended work and that hazardous biological or chemical work is not being performed in an unsuitable unit.
3. Check the certification or maintenance status and confirm that the airflow system is operating within its approved specifications.
4. Inspect the work surface, front opening, grille, glass panels, and power cable for damage or obstruction.
5. Remove unnecessary items and disinfect the work surface using a compatible approved disinfectant.
6. Wear the required laboratory PPE and arrange materials before starting work.
7. If UV lighting is fitted, ensure it is switched off before work and follow the manufacturer's UV safety instructions.

How to use

1. Switch on the blower and allow the cabinet to stabilise for the time specified by the manufacturer.
2. Check airflow or filter indicators and report any alarm or abnormal reading before proceeding.
3. Disinfect the work surface and allow the disinfectant to dry as required.

4. Place only the materials needed for the procedure inside the work zone without blocking the airflow grille.
5. Arrange clean and used materials to reduce cross-contamination and avoid unnecessary movement across the airflow path.
6. Perform the approved procedure using slow, controlled movements and keep hands and equipment within the designated work area.
7. Keep the front opening unobstructed and avoid rapid movements that disturb airflow.
8. Monitor the cabinet for unusual noise, airflow changes, or warning indicators throughout the task.
9. Remove materials safely and disinfect the work surface after the procedure.
10. Keep the blower running for the specified purge period if required by the cabinet procedure.
11. Switch off the cabinet according to the approved shutdown procedure and record any faults.

⚠ Safety

- Do not use a laminar air flow cabinet as a substitute for a biological safety cabinet.
- Do not handle infectious agents, hazardous chemicals, or toxic vapours unless the cabinet is specifically designed and approved for that purpose.
- Never block the airflow grille or place large items where they disrupt the filtered airflow.
- Never look directly at or expose skin to an operating UV lamp.
- Do not work in the cabinet if airflow alarms, filter faults, or certification issues are unresolved.
- Follow laboratory biosafety, chemical safety, and waste-disposal procedures for all materials.

Care & storage

- Clean and disinfect the work surface after each session using compatible products.
- Keep air grilles and intake areas free of paper, packaging, and debris.
- Check the blower, lights, controls, airflow indicators, and visible seals regularly.
- Arrange scheduled certification, HEPA filter testing, and maintenance through authorised personnel.
- Report abnormal airflow, filter alarms, or cabinet damage and do not use until cleared.

Common mistakes to avoid

- Using a laminar flow cabinet for hazardous work that requires operator protection.
- Blocking the airflow path with equipment or materials.
- Starting work before the airflow stabilises or after an unresolved alarm.
- Using UV lighting while personnel are exposed to direct UV radiation.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.